Where Is the Speaker? Silence-Informed Localization, Distance Cues and Selective Beamforming on a 4-Mic ReSpeaker

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Abstract

Service robots often hear through a small circular microphone array but use it as a single microphone. We study what the four raw microphones of a ReSpeaker USB v2.0 (6.4 cm aperture) can add to a robot's speech pipeline: the talker's angle, the talker's distance, and the suppression of sound from other directions before neural denoising. The common thread is a *silence model*: the pitch-based voice activity detector (VAD) the pipeline already runs splits the stream into voice and silence, and the silence frames give the cross-spectra of everything that is not the talker. In image-source simulations with a voiced talker at 0 dB SNR and a directional fan, plain SRP-PHAT has a median per-frame error of 35.0/78.9° (RT_{60} 0.3/0.6 s); subtracting the silence cross-spectra and weighting bins by speech presence brings it to 4.5/4.3°, and a streaming tracker follows a moving talker with a median error of 0.7°. Distance cannot come from time differences on this aperture; the coherent-to-diffuse ratio is the useful cue, but it is room-relative: cross-room error is 0.53 m without and 0.33 m with the room's RT_{60} (predicting the mean: 0.75 m). An MVDR beamformer steered at the estimated talker improves output SINR against a competing talker by 7.4 dB on average. A first hardware session in a real room changes two things. The raw microphones carry low-frequency hum that the simulation lacked (65 % of the noise power below 100 Hz); it broke the VAD (9 speech frames in 26 s of talking) until its input was high-passed at 200 Hz. The simulation-trained distance model does not transfer, so we replace it with a microphone-only near/mid/far estimate that learns its reference from the session's own utterances; it separates near from far talkers but moves with vocal effort. Code, live radar and hardware tools are released.

1 Introduction

Robots that take spoken commands in homes and competitions face the cocktail-party problem with very little hardware. A common choice is the ReSpeaker USB Mic Array v2.0: four omnidirectional microphones on a 32 mm-radius circle with an on-chip DSP that outputs one processed channel and a firmware direction-of-arrival (DOA) estimate. Our robot's speech pipeline uses exactly that processed channel: it feeds a neural denoiser (DeepFilterNet3 [9]), then a voice activity detector (VAD) built from an energy gate, an autocorrelation pitch test in 85–255 Hz and an MFCC speaker lock, then speech recognition. Spatial information is discarded, and the published firmware DOA is smoothed with a linear average, which maps a talker crossing 0° to 180°.

We ask what the raw microphones can add, in three steps a robot needs: *where* the talker is (angle), *how far* (distance), and *attending* to them by suppressing other directions before the denoiser. Our organising idea is that the VAD already provides the most useful reference signal: frames without voice describe the room – fans, motors, diffuse noise – and their cross-spectra can be used to clean the localisation and to train the spatial filter.

Contributions: (i) a streaming localiser (pitch VAD → silence model → SRP-PHAT → distance features → wrap-safe tracker) with an ablation of each use of the silence model; (ii) an analysis of which acoustic cues give distance on a 6.4 cm array, showing that the coherent-to-diffuse ratio works within a room and that a room covariate (RT60) is what makes it transfer; (iii) a measurement of MVDR versus delay-and-sum selectivity on this aperture with an estimated, not oracle, steering direction; (iv) a first hardware session in a real room, which shows that low-frequency hum on the raw microphones breaks the VAD (fixed by high-passing its input) and that the simulation-trained distance model does not transfer; (v) a microphone-only near/mid/far estimate that learns its reference online instead of being calibrated per room; (vi) tools for the hardware study (live radar, labelled recording, calibration, session analysis) and an integration plan for a ROS 2 pipeline. The angle, distance-regression and beamforming numbers are from simulation; the hardware results are a first, partly unlabelled session.

2 Related work

DOA with small arrays. GCC-PHAT [6] and steered-response power with PHAT weighting (SRP-PHAT) [3] are the standard robust estimators under reverberation; ODAS [4] runs SRP-PHAT with tracking on the same ReSpeaker and similar arrays for robots. Speech-presence or SNR-dependent weighting of time–frequency bins is a known improvement for GCC-type methods in noise; we evaluate it, and noise cross-spectrum subtraction, as two uses of one silence model.

Distance estimation. Source distance from a compact array relies on the direct-to-reverberant ratio (DRR), level, or learned features [11]. The coherent-to-diffuse ratio (CDR) is a coherence-based DRR proxy with closed-form estimators [10]; it depends on the room's critical distance, which scales with $\sqrt{V/\mathrm{RT}_{60}}$ [7].

Spatial filtering before enhancement. MVDR beamforming [2] followed by single-channel enhancement is the classic multichannel speech-enhancement chain; small-aperture arrays need superdirective designs and are sensitive to steering errors. Neural single-channel denoisers such as DeepFilterNet3 [9] remove stationary and diffuse noise but cannot separate two talkers by direction.

Simulation. We use the image-source method [1] via pyroomacoustics [8], and a diffuse noise field with the exact spherical-isotropic coherence [5].

3 Method

Signal model. Four microphones p_m on a 32 mm circle, 16 kHz, 512-point Hann STFT, hop 256. A far-field source at azimuth θ gives $X_m(f) = S(f)e^{-j2\pi f\tau_m(\theta)}$, $\tau_m = -p_m^\top u(\theta)/c$. The largest inter-microphone delay is 0.19 ms (≈3 samples).

VAD and silence model. Per frame, the mean of the four channels over the last 1024 samples, high-passed at 200 Hz (fourth-order Butterworth, zero phase; added after the hardware session, Section 4.1), passes an energy gate and an autocorrelation test: the normalised autocorrelation peak in the lag band of 85–255 Hz must exceed 0.6 (the robot pipeline's rule), which also yields F_0. The gate is $\max(g_0, \phi \cdot 10^{6/20})$ where ϕ follows the RMS of non-speech frames (fast down, slow up), because the raw microphones are not gain-controlled. Non-speech frames, after a 0.3 s hangover, update the noise cross-spectral matrix Φ_n (time constant 2 s). Speech frames update Φ_x^{fast} (0.25 s) and Φ_x^{slow} (1 s); both are reset when speech resumes after 0.6 s of silence.

Angle. SRP-PHAT on a 1° grid in 200–3800 Hz:

$$P(\theta) \propto \sum_{i<j} \sum_f w_f \,\mathrm{Re}\Big\{ \tfrac{R_{ij}(f)}{|R_{ij}(f)|} e^{j2\pi f(\tau_i(\theta)-\tau_j(\theta))} \Big\},$$

with $R = \Phi_x^{\mathrm{fast}} - \Phi_n$ (noise subtraction) and $w_f = \max(0, 1 - \mathrm{tr}\,\Phi_n/\mathrm{tr}\,\Phi_x)$ (speech-presence weight). Optional: a Gaussian comb at multiples of F_0. The peak is refined parabolically; confidence is the peak's height above the map mean. Baseline: per-pair GCC-PHAT upsampled 16× and a least-squares fit of $u(\theta)$.

Distance. From $\Phi_s = \Phi_x^{\mathrm{slow}} - \Phi_n$: (a) CDR with the DOA-independent estimator of [10] per pair, using bins where the diffuse coherence $\mathrm{sinc}(2\pi f d/c) < 0.7$ in 0.4–7 kHz, median over bins and over the six pairs; (b) speech level (dBFS); (c) SNR over Φ_n; (d) spectral tilt. A ridge regression maps standardised features to $\log r$; RT_{60} can be appended as a room covariate (the ReSpeaker firmware reports an estimate).

Relative range. Without a room model the array can only say whether the talker is nearer or farther than usual. Per speech frame with SNR above 3 dB we take a cue x: the noise-subtracted speech power $x = 10\log_{10}\mathrm{tr}_{200\text{–}3800\,\mathrm{Hz}}\Phi_s$ (default) or the CDR. Each utterance (closed by 0.6 s of silence) adds the median of its x to a history of the last 20. With c the history's median and $w = \max(3\,\mathrm{dB}, 1.48\,\mathrm{MAD})$, the running median $\bar{x}$ of the current utterance is *near* if $\bar{x} > c + w$, *far* if $\bar{x} < c - w$, *mid* otherwise, with 0.5 dB hysteresis. Until three utterances exist the history is padded with a hardware prior (normal speech at 1 m on the raw, fixed-gain microphones). The noise reference is the continuously re-learned silence model; nothing is calibrated per room.

Tracking. An alpha-beta filter on the circle (residuals wrapped to $[-180°, 180°)$) and an EMA of $\log r$; measurements below a confidence threshold are ignored, and tracks expire after 3 s.

Selective filtering. MVDR $w = (\Phi_n + \delta I)^{-1}a/(a^H(\Phi_n + \delta I)^{-1}a)$ steered at the estimated azimuth, $\delta = 1\%$ of per-bin power, against delay-and-sum $w = a/4$. In the beamforming experiment the target's azimuth is estimated by SRP-PHAT on the difference between cross-spectra after and before the target's onset, and Φ_n is the pre-onset covariance (interferer plus noise).

Simulation. A shoebox room (8×7×3 m, image-source order ≤25, Sabine absorption for the target RT_{60}), array at 1.1 m height placed off-centre, mouth at 1.5 m. The talker is a synthetic voiced signal (random F_0 in 100–220 Hz, vowel formants, syllabic envelope, pauses), −30 dBFS at 1 m with ±4 dB effort jitter in the distance sweep. Noise: spherically diffuse pink noise [5] plus mic

DOA	8 angles × SNR {0,10,20} × RT_{60} {0.3,0.6} × {diffuse, fan}
Distance	6 distances 0.5–3 m × 3 angles × RT_{60} {0.3,0.45,0.6} × 2
Beam	interferer at 45–180°, SIR 0 dB, RT_{60} {0.3,0.6}
Demo	7 positions, 0.5–3 m, RT_{60} 0.45, 17.5 s

Table 1: Local simulation sweeps (`local_small.yaml`).

VAD high-pass	diff. 0.3	diff. 0.6	fan 0.3	fan 0.6
none	3.2	3.0	4.2	2.6
100 Hz	1.8	3.2	3.0	3.2
150 Hz	1.9	2.8	5.1	11.8
200 Hz	2.1	15.0	5.8	14.1

Table 2: Simulation: DOA MAE (deg) of the full method at 0 dB SNR versus the VAD high-pass cut-off (columns: noise, RT_{60}). The simulated noise has no hum; the filter only removes F_0 energy and lets one reverberant clip fail at 150–200 Hz (max errors 73 and 60°).

self-noise, optionally a directional pink "fan" at 200°, 2 m. Sweeps are listed in Table 1.

4 Results

Angle. Figure 1. At 10–20 dB SNR the full method is within 2°. At 0 dB the two uses of the silence model matter in different conditions (pooled per-clip MAE, RT_{60} 0.3/0.6 s; per-frame medians in brackets). With diffuse noise, dropping the speech-presence weights raises the error from 2.1/15.0° (2.6/2.7) to 21.9/35.3°: after subtraction, noise-only bins carry structured residue that PHAT normalises to full weight. With the fan, plain SRP-PHAT locks on the fan (42.9/56.4°, medians 35.0/78.9), while the full method keeps a median of 4.5/4.3°. The MAE of 15.0/14.1° at RT_{60} 0.6 s comes from a single clip per cell, and from the VAD high-pass added for the hardware (Table 2): without it the same cells were 3.0/2.6°. GCC-PHAT with least squares reaches 19.0–51.9° at 0 dB. The F_0-harmonic comb changes nothing measurable on our synthetic talker, whose spectrum is already purely harmonic. At 0 dB, 7 of the 8 clips per cell have any VAD-speech frame.

Tracking a moving talker. Figure 2: after 0.5 s of settling per position, the tracked azimuth has a median error of 0.7° (90th percentile 7.1°). The distance track (model trained in the same room) has a median error of 0.13 m and under-estimates the 3 m position. The relative near/mid/far estimate (no room model) gets the modal band right at 4 of 7 positions (51 % of frames): right at 1 m and 3 m, wrong in between, where the speech level changes by only a few dB.

VAD input	speech while talking	"speech" in silence
broadband	12/477	0/68
150 Hz high-pass	134/477	13/68
200 Hz high-pass	135/477	0/68

Table 3: Hardware VAD on the labelled clip (frames).

Distance. Figure 3: the CDR is the strongest cue ($\rho = -0.77$ pooled, -0.90 to -0.95 within a room). Level ($\rho = -0.62$) suffers from the ±4 dB vocal-effort jitter, SNR (−0.44) from how much speech leaks into the noise estimate, and tilt (+0.08) carries little information. Figure 4: across angles, CDR alone gives 0.44 m MAE and CDR + level 0.42 m, against 0.75 m for predicting the mean and 0.92 m for an inverse-square law on level. Adding the room's RT_{60} gives 0.24 m, and across rooms (leave one RT_{60} out) 0.33 m versus 0.53 m without it.

Selective cancellation. Figure 5: MVDR gains 2.6–10.8 dB (mean 7.4 dB), less in the more reverberant room and at 45° separation, where the estimated target DOA was also furthest off (9.6°). Delay-and-sum gains 0.2 dB: at 1 kHz the array spans 0.19 wavelengths, so only superdirective weights give selectivity.

4.1 Hardware session

A ReSpeaker USB v2.0 (6-channel firmware) on a laptop in a home office; the firmware estimates $RT_{60} = 0.9$ s, beyond the simulated rooms. Data: one labelled clip (front, 0.5 m), a short live log with the simulation's VAD and an 18-minute live log with the final configuration, in which the talker moved between about 1 m, close and about 3 m without labels.

The VAD failed first. The noise floor is -41.8 dBFS with 65 % of its power below 100 Hz (Figure 6, left); speech raises the broadband level by less than 1 dB. With the simulation's VAD, 9 of 1645 frames were speech during 26 s of talking, so the radar never produced a distance. The angle from those few frames was stable: SRP-PHAT works in 200–3800 Hz, above the hum. Table 3: a high-pass on the VAD input restores coverage; at 150 Hz the hum's harmonics still pass the pitch test in silence, at 200 Hz they do not. With 200 Hz, 22 % of the long session's frames are voiced speech.

The distance model does not transfer. On the 0.5 m clip the CDR is -3.2 dB (simulation at 0.5 m: about +7 to +10 dB) at an SNR of 10.6 dB, and the simulation-trained model reads 2.5 m. This is the room dependence of Figure 3 in a more reverberant room than any simulated one, plus a quieter talker.

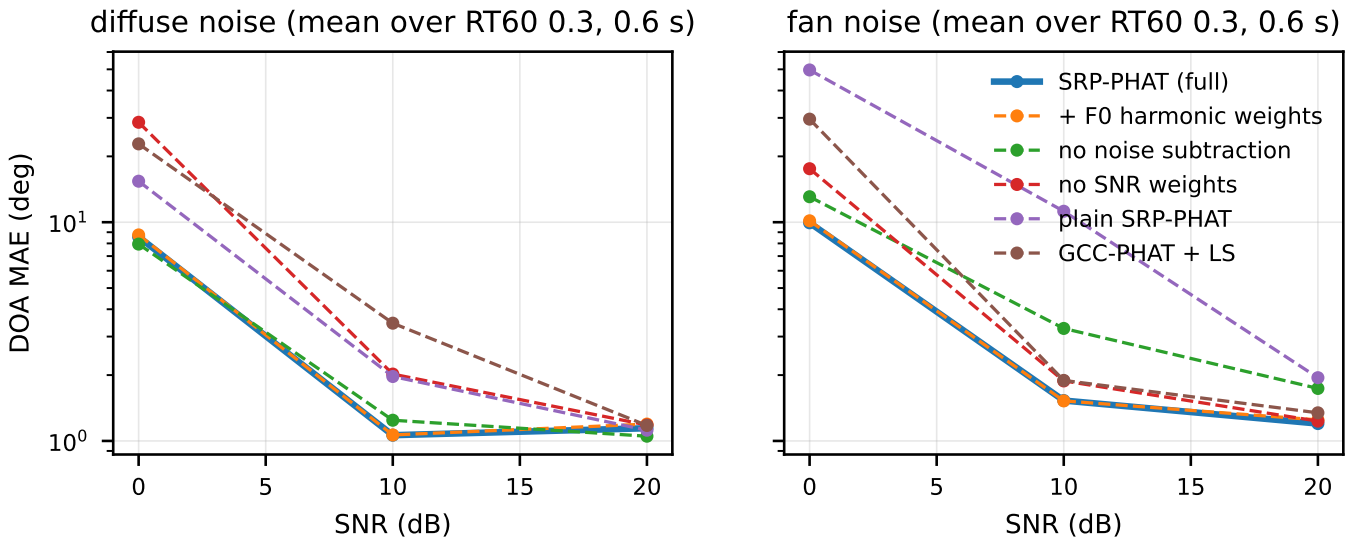


Figure 1: DOA mean absolute error (log scale) versus SNR, averaged over RT_{60} 0.3 and 0.6 s, talker at 1.5 m. Removing either use of the silence model hurts in one noise condition; removing both fails with a directional noise source.

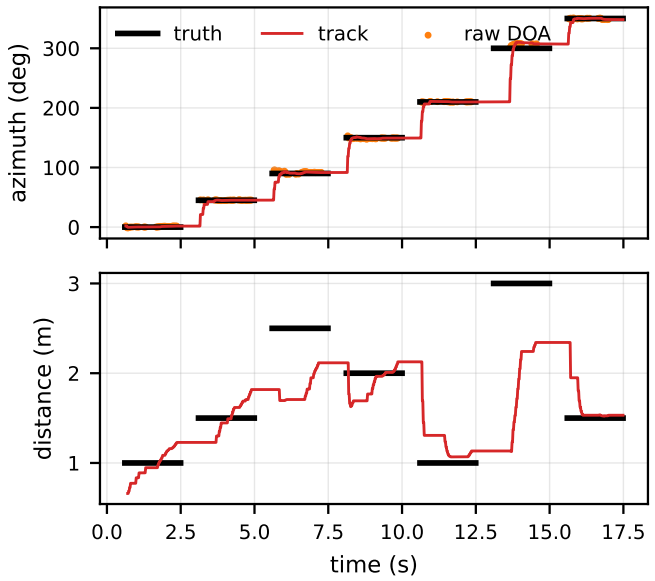


Figure 2: Streaming estimates for a talker moving through 7 positions (black: truth). Top: azimuth. Bottom: distance.

Online near/mid/far. In the long session the speech-level estimate spent 14/74/12 % of utterances near/mid/far and changed band between consecutive utterances 28 % of the time. Replaying the same utterances with the CDR as the cue gives 20/69/11 % and 18 % changes. The two cues agree on 59 % of utterances and are only weakly related (Spearman 0.28; Figure 6, right). In use, near and far were told apart, and speaking louder or softer moved the band. Without distance labels we cannot say which cue is right more often.

5 Discussion

The silence model is the method. The estimators are textbook. What made them work on this array was getting "noise" right: which bins contain it, which frames it can be learned from (a fixed energy gate tuned for the processed channel missed a talker at 3 m entirely), and when old statistics stop describing the scene (a talker who moved between utterances). The hardware session made the same point a fourth time: the real room's hum, absent from the simulation, decided which frames were "voice", and every downstream estimate inherits that decision. A high-pass on the VAD input fixed it on hardware and cost one reverberant clip in simulation; the cut-off is a property of the deployment's noise, not of the method.

Distance is a room-calibrated quantity. A 6.4 cm array has no geometric range resolution; the CDR measures distance relative to the room's critical distance. Simulation predicts and the cross-room experiment confirms that a room descriptor is needed, and the real room was outside the simulated range. A robot that changes rooms cannot be calibrated per room, so the microphone-only output we keep is relative: near, mid or far compared with what the robot has heard in this session. This is honest about what one small array knows, but it inherits the cue's confounds. The speech level confuses distance with vocal effort; the CDR does not depend on effort but on the room and on SNR. In simulation, 1 m to 3 m spans only about 7.5 dB of speech level, so three bands are separable only for distance changes of about a factor two. Other sensors are the natural source of the missing reference: a person detector with depth gives distance labels whenever it sees the talker.

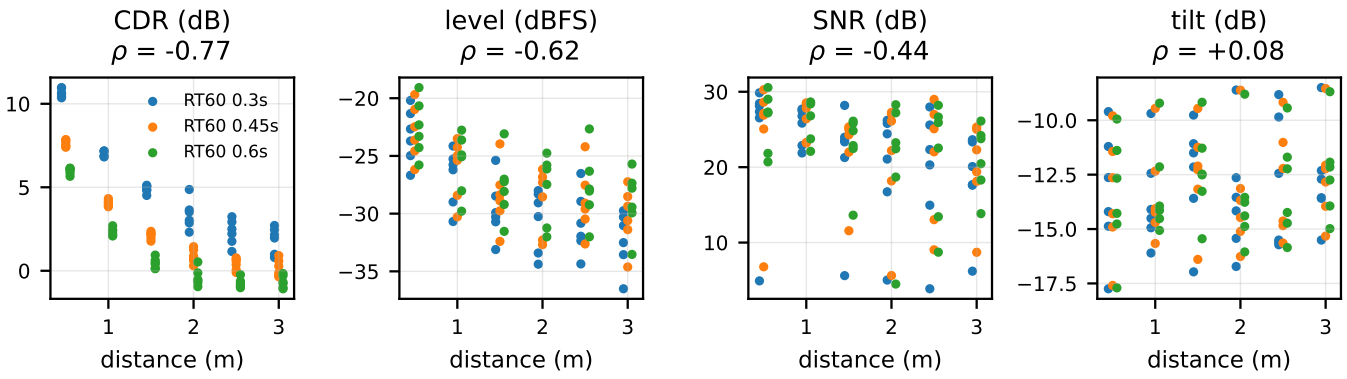


Figure 3: Distance features versus true distance (108 clips), coloured by RT_{60}; ρ is Spearman's correlation with $\log r$ over all rooms. CDR is monotone within each room but offset between rooms.

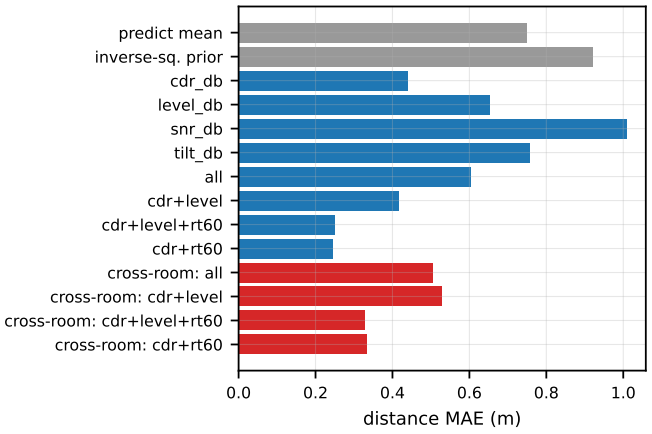


Figure 4: Distance MAE. Blue: leave-one-angle-out; red: leave-one-RT_{60}-out.

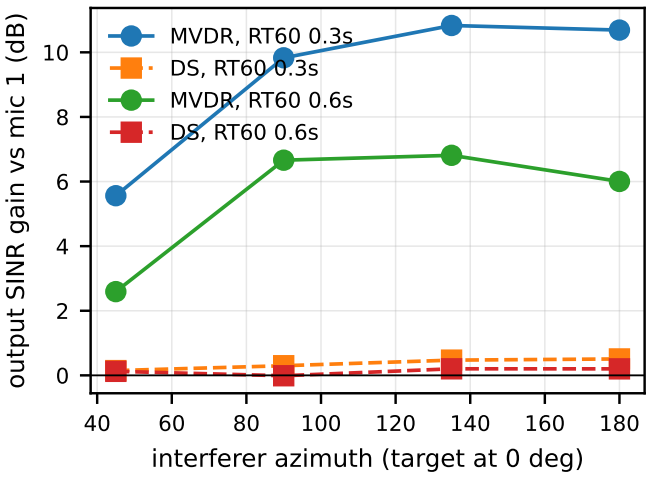


Figure 5: Output SINR gain over microphone 1 against a competing talker (target at 0°, both at 1.2 m, SIR 0 dB), with steering from the estimated DOA.

Where simulation is optimistic. The synthetic talker is more periodic and spectrally simpler than speech, which helps both the VAD and PHAT; the shoebox image-source model has specular reflections and no furniture, scattering or robot body; the diffuse field is ideal; there is no low-frequency hum; and the microphones are perfectly matched and positioned. The first two showed on hardware: the room was more reverberant than any simulated one and the noise was hum-dominated.

Limitations. Small simulation sweeps (8 clips per DOA cell, 108 distance clips, 8 beamforming scenes) without confidence intervals; a larger configuration is provided. The hardware session has one labelled clip and one unlabelled live session from one talker in one room: the labelled angle grid, the orientation calibration and therefore the angle error and the comparison with the firmware DOA, labelled distances to choose between the cues, and a two-talker beamforming test are still to be done (the tools and protocol are in the repository). One interferer, equal levels; no moving interferer; no own-voice (TTS) handling.

6 Conclusion

On a four-microphone, 6.4 cm array, the voice/silence split a robot's speech pipeline already computes is enough to turn the raw channels into a usable sense of where the talker is, provided the split is right. In simulation: a few degrees of median angle error at 0 dB SNR when the silence frames are used both to subtract noise and to weight frequency bins, distance within about a third of a metre once the room's reverberation is known, and several dB of suppression of a competing talker with an MVDR filter steered at the estimate. On hardware, the direction followed the talker once the VAD was protected from the room's hum, while metric distance did not transfer to an unseen room; a relative near/mid/far estimate that learns from the session itself is what the microphones alone can offer. Next steps are the labelled hardware grid (angle error against the firmware DOA, and labelled distances to choose the range cue), a two-talker beamforming test, and integration as ROS 2 nodes ahead of the existing DeepFilterNet3 denoiser, where a person detector can supply the distance reference the array lacks.

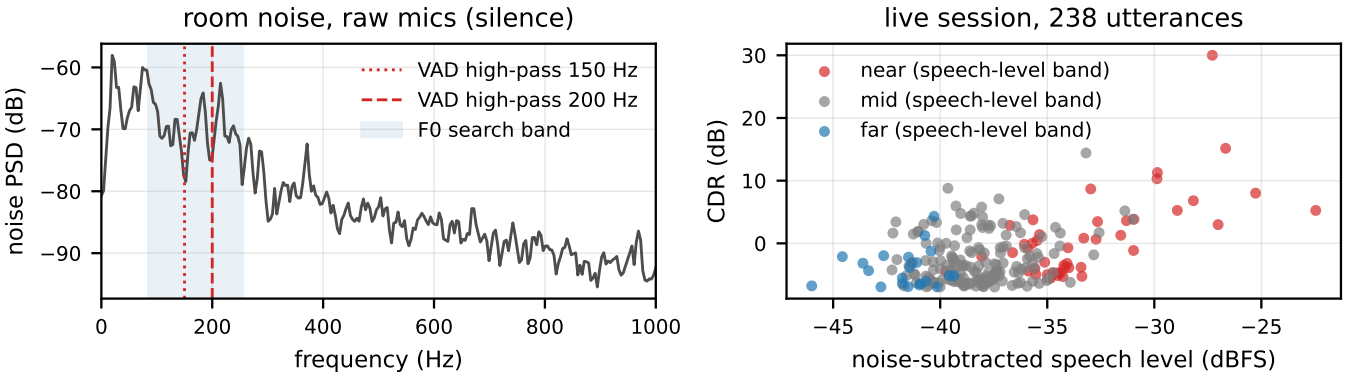


Figure 6: Hardware session. Left: noise spectrum on the raw microphones in a real room (silence of a labelled clip), with the VAD's F_0 search band and the two high-pass cut-offs tried. Right: per-utterance cues in an 18-minute live session (238 utterances), coloured by the band the online speech-level estimate assigned.

Reproducibility. `make all` in `projects/15_spatial_audio_attention` regenerates every simulation number and figure in about four minutes on a laptop CPU; `make hw` regenerates the hardware numbers from the session's raw logs (kept locally; the derived JSON is in the repository), and `make live`, `make record` and `make calibrate` run the hardware study.

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